

SahiRate (■■■■ ■■■■)

Master Project Manual & Technical Architecture Guide

Smart India Hackathon (SIH) Top-5 Final Evaluation Blueprint
Ministry of Mines (MoM) / JNARDDC (Nagpur)

1. Executive Summary & SIH Problem Context

Problem Statement: 'Kabadiwala Connect – Bringing the Informal Collector into the Formal Recycling Chain' under the Ministry of Mines and JNARDDC (Nagpur).

- **Ground Reality:** 95% of e-waste is captured by informal waste-pickers due to extensive last-mile reach.
- **Middleman Exploitation:** Rigged scales (-25%) and arbitrary cuts (-20%) leave collectors with only ■48/kg on ■80/kg benchmark commodities.
- **Backyard Processing Hazards:** Open-air cable burning (dioxins) and acid leaching contaminate local soil and groundwater.
- **SahiRate Solution:** Gives informal collectors offline AI tools, multilingual audio prices, verified scales (+66% net earnings), and direct access to CPCB-registered yards.

2. System Architecture & Components

Subsystem	Technology Stack	Core Functionality
Collector App (frontend/)	React 18, Vite, TypeScript, PWA/TWA, Dexie DB, ONNX WASM	Offline AI classification, spoken audio price board, lot creation, and QR handover.
Recycler & Admin Portal (frontendRA/)	React 18, Vite, TypeScript, Tailwind, SVG Charts, CSV/JSON Exporter	Scrap intake, digital weighment inspection, PCB KYC approval, unit economics simulator, 6 datasets.
Central Backend (backend/)	FastAPI, Python 3.13, SQLAlchemy, SQLite, Pytest, JWT Auth	Idempotent outbox sync, role-based auth, transaction ledgers, audit trail.

3. Core Feature Catalog & Implementation

A. Edge AI Material Scanner (ONNX Runtime WASM)

Runs 100% offline computer vision classification into 7 scrap categories directly in browser memory without internet.

B. JNARDDC Critical Minerals & Rare Earths Tracker

Formulates exact recovery of **Neodymium (Nd ~65g)**, **Dysprosium (Dy ~12g)** from motors, **Tantalum (Ta ~45mg)** from PCBs, **Lithium (Li ~85g)** and **Cobalt (Co ~140g)** from batteries. 1 ton of e-waste yields 40× more gold and 25× more copper than 1 ton of mined ore.

C. 7-Day Value History & Spoken Audio Price Board

Displays daily CPCB rates with interactive 7-day SVG trajectory charts. Uses Web Speech Synthesis in Hindi, Marathi, Bengali, Odia, and English for low-literacy inclusion.

D. Deterministic Offline/Online Sync Simulation

Allows evaluators to toggle 'Sync OFF' to verify pure Dexie local DB lot creation, then flip to 'Sync ON' to observe automatic ingestion into FastAPI POST /api/v1/sync/events.

E. Unit Economics & 1.5% EPR Financial Sustainability

Collector nets ₹80/kg on SahiRate vs ₹48/kg with middlemen (+66.7% income lift). Zero fees charged to collectors; funded by a 1.5% EPR transaction fee paid by smelters (₹1,200/metric ton).

F. The 6 Standardized Circular Datasets & 1-Click Export Center

Accessible at /admin/datasets with 1-click RFC-4180 CSV and JSON downloads for Material Master, Daily Prices, Recyclers, Transactions, Traceability, and Collector Impact.

4. Multilingual Translation Audit

Language	Code	Speech Synthesis Accent	Status
English	en	en-IN (Indian English)	VERIFIED (100% Populated)
Hindi (■■■■■■■)	hi	hi-IN (Devanagari)	VERIFIED (100% Populated)
Marathi (■■■■■■■)	mr	mr-IN (Maharashtra)	VERIFIED (100% Populated)
Bengali (■■■■■■■)	bn	bn-IN (West Bengal)	VERIFIED (100% Populated)
Odia (■■■■■■■)	or	or-IN (Odisha)	VERIFIED (100% Populated)

5. Operational Runbook & Credentials

- **Backend:** cd backend -> .\env\Scripts\activate -> uvicorn app.main:app --reload --port 8000 (http://localhost:8000)
- **Collector App:** cd frontend -> npm run dev (http://localhost:5173)
- **Recycler & Admin Portal:** cd frontendRA -> npm run dev -- --port 5174 (http://localhost:5174)
- **Collector Demo Login:** collector@sahirate.in / sahirate123 (Ramesh Kumar • Ward 14, Nagpur)
- **Recycler Demo Login:** recycler@ecorecycle.in / sahirate123 (EcoRecycle Yard Ltd)
- **Admin Demo Login:** admin@sahirate.in / sahirate123 (Super Admin)